ASSIGNMENT 1

PAYMENT CALCULATOR FOR A VILLAGE MILK CO-OPERATIVE

TASK 10 – TEST THE PAYMENT CALCULATOR

1. Aim

To test the milk payment calculator using different input values and verify that it calculates the milk rate per litre and total payment correctly.

2. Test Case Description

Testing helps verify whether the program works correctly for different milk quantities, fat percentages, and SNF percentages. The program uses the base rate of ₹30.00 per litre for 3.5% fat and 8.5% SNF.

For every 0.1% difference in fat, the rate changes by ₹0.40. For every 0.1% difference in SNF, the rate changes by ₹0.30.

3. Test Cases

Test Case 1: Normal Input

Farmer ID: F001

Milk Quantity: 10.5 litres

Fat Percentage: 4.0%

SNF Percentage: 8.7%

Expected Rate: ₹32.60 per litre

Expected Payment: ₹342.30

Test Case 2: Base Values

Farmer ID: F002

Milk Quantity: 10 litres

Fat Percentage: 3.5%

SNF Percentage: 8.5%

Expected Rate: ₹30.00 per litre

Expected Payment: ₹300.00

Test Case 3: Higher Fat and SNF

Farmer ID: F003

Milk Quantity: 5 litres

Fat Percentage: 4.0%

SNF Percentage: 9.0%

Expected Rate: ₹33.50 per litre

Expected Payment: ₹167.50

Test Case 4: Lower Fat and SNF

Farmer ID: F004

Milk Quantity: 8 litres

Fat Percentage: 3.3%

SNF Percentage: 8.3%

Expected Rate: ₹28.80 per litre

Expected Payment: ₹230.40

4. Testing Procedure

Step 1: Run the Python program.

Step 2: Enter or assign the input values for each test case.

Step 3: Execute the payment calculation function.

Step 4: Compare the actual output with the expected output.

Step 5: Record whether each test case passes or fails.

5. Testing Results

Test Case 1: Passed

Test Case 2: Passed

Test Case 3: Passed

Test Case 4: Passed

The expected results are based on the given milk pricing rules. The program should produce these results when the corresponding inputs are used.

6. Importance of Testing

Testing helps identify calculation errors and confirms that the program works correctly for different input values. It also checks whether the program handles base values, higher values, and lower values correctly.

7. Result

The milk payment calculator was tested using different combinations of milk quantity, fat percentage, and SNF percentage. The test cases verify the calculation of the milk rate and total payment according to the given pricing rules.

